

## Exercise 6

### Date Functions

Table 1: Employees

| emp-id | name    | hire-date  | hire-plus<br>6-months |
|--------|---------|------------|-----------------------|
| 1      | Alice   | 2020-01-15 | 2020-07-15            |
| 2      | Bob     | 2021-06-10 | 2021-12-10            |
| 3      | Charlie | 2023-03-22 | 2023-09-22            |

```
SELECT
    emp-id,
    name,
    hire-date,
    DATEADD(Month, 6, hire-
    plus-6-months
FROM Employees;
```

Table 2: Students

| Student-id | Name   | age-in-days          |
|------------|--------|----------------------|
| 101        | Marya  | (Today - 2005-08-10) |
| 102        | Ethan  | (Today - 2004-12-01) |
| 103        | Sienna | (Today - 2006-03-15) |

```
Select
    Student-id,
    Name,
    DateDiff(Day, dob,
    (cast (GetDate()) AS
    Date)) AS
    age-in-days
FROM Students;
```

Table 3: Invoices

| Invoice-id | Issue-date | due-date   | days-between |
|------------|------------|------------|--------------|
| 501        | 2025-03-10 | 2025-03-25 | 15           |
| 502        | 2025-04-1  | 2025-04-15 | 14           |
| 503        | 2025-04-10 | 2025-04-20 | 10           |

```
Select
    invoice-id,
    Issue-date,
    due-date,
    DATEDIFF(DAY, Issue-date,
    due-date) AS days-between
FROM Invoices
```

Table 5: Courses

| Course-id | Name       | Formatted-date |
|-----------|------------|----------------|
| 201       | SQL Basics | May 2025       |
| 202       | Python     | June 2025      |

SELECT

Course-id,  
name,  
TO-CHAR (start-date,  
Month 'YY') AS  
Formatted-date  
FROM courses;

Table 6: Membership

| Member-id | Full-start-date |
|-----------|-----------------|
| 1         | 2023-05-10      |
| 2         | 2022-11-25      |

Select

member-id,  
DATE-FROM-PARTS (start  
year, start-month, start  
day) AS  
Full-start-date  
FROM membership;

Table 7: Subscriptions

| Sub-id | plan    | extended-renewal-date |
|--------|---------|-----------------------|
| 11     | Basic   | 2026-01-01            |
| 12     | Premium | 2026-03-15            |

Select

sub-id,  
plan,  
DATEADD (year, 1,  
renewal-date) AS  
extended-renewal-date  
FROM subscriptions;

Table 8 : Orders

| Order_id | Order_date | Today_date | days-since-order     |
|----------|------------|------------|----------------------|
| 1001     | 2025-04-15 | Today      | (Today - 2025-04-15) |
| 1002     | 2025-04-10 | Today      | (Today - 2025-04-10) |

Select  
order\_id,  
Order\_date,  
Current-Date AS today-  
date,  
DATE DIFF ( day,  
order\_date /  
Current-Date) AS  
days-since-order  
From Orders;

Table 9: Trainings

| training_id | topic      | training_year |
|-------------|------------|---------------|
| 1           | Safety     | 2025          |
| 2           | Compliance | 2025          |

Select  
training\_id,  
topic,  
EXTRACT (YEAR FROM  
training-date) AS  
training\_year  
From Trainings;

Table 10 : Blog\_Posts

| Post_id | title         | hour_published | minute_published |
|---------|---------------|----------------|------------------|
| 1       | SQL Tips      | 10             | 15               |
| 2       | Data Cleaning | 10             | 45               |

Post-id,  
title,  
Extract (Hour FROM  
Published-on) AS  
hour\_published;  
EXTRACT (minute  
FROM published-on,  
AS minute-publish  
From Blog\_Posts;

Table 11: Drivers

| driver-id | License-expiry | days-left        |
|-----------|----------------|------------------|
| 301       | 2025-08-10     | (Expiring-Today) |
| 302       | 2025-12-31     | (Expiring-Today) |

Select

driver-id,  
license-expiry,  
DATEDIFF (day, Current-DATE,  
license-expiry) AS days-left  
FROM Drivers;

Table 12: Messages

| Message-id | Sent-timestamp         | Current-timestamp | Seconds-since-sent     |
|------------|------------------------|-------------------|------------------------|
| 1          | 2025-04-19<br>09:32:45 | NOW               | (NOW - Sent-timestamp) |
| 2          | 2025-04-18<br>23:59:59 | NOW               | (NOW - Sent-timestamp) |

Select

message-id,  
Sent-timestamp,  
Current-Timestamp AS  
Current-timestamp,  
DATEDIFF (second,  
Sent-timestamp,  
CURRENT-Timestamp) AS  
Seconds-since-sent  
FROM Messages;

Table 13: Returns

| Return-id | Return-date | Restock-date |
|-----------|-------------|--------------|
| 901       | 2025-04-05  | 2025-04-20   |
| 902       | 2025-04-01  | 2025-04-16   |

Select

return-id,  
return-date,  
DATEADD (day, 15, return-  
date) AS restock-date  
FROM Returns;

Table 14: Assignments

| Assign-id | Assigned-on-date |
|-----------|------------------|
| 1         | 2025-03-01       |
| 2         | 2025-03-05       |

Select

Assign-id,  
TO DATE (assigned-on) AS  
assigned-on-date  
FROM Assignments;

Table 15: Meetings

| Meeting-id | Formatted-meeting-time    |
|------------|---------------------------|
| 1          | April 19, 2025 at 2:00PM  |
| 2          | April 19, 2025 at 9:30 AM |

Select  
meeting-id,  
TO-CHAR(  
Scheduled-time,  
'Month DD, YYYY "at"  
HH:MM AM/PM')  
AS formatted-meeting-  
time  
from Meetings;